

An Immersive Serve Type Recognition Skill Training System for Table Tennis Beginners

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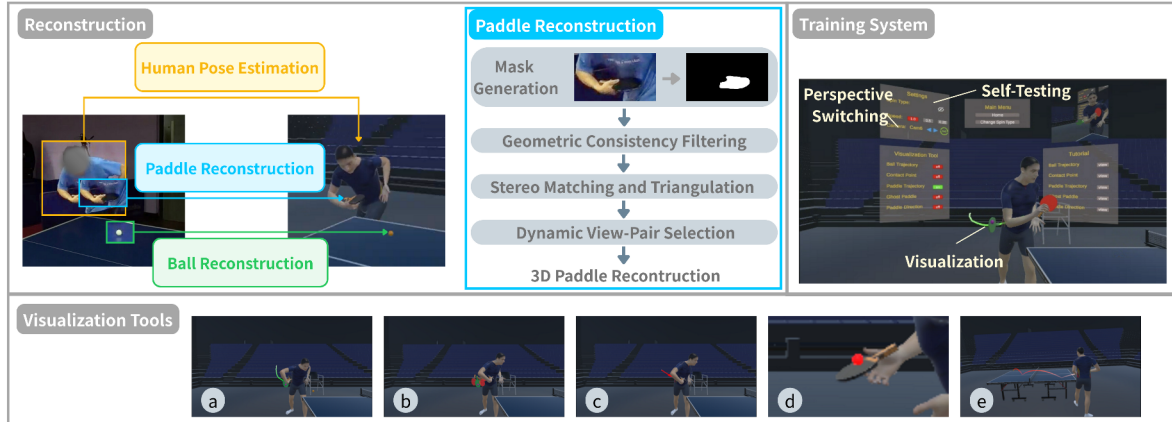


Figure 1: We present an immersive serve type recognition training system designed for table tennis beginners. The reconstruction module extracts key motion data from multi-view serving videos and feeds it into an immersive training environment that supports interactive learning with visualization and self-testing features. The system provides five visualization tools to support learning by highlighting key observation points related to the serve: (a) Paddle trajectory illustrates the paddle's motion path. (b) Ghost paddles show intermediate paddle positions. (c) Paddle direction visualizes swing direction using arrows. (d) Contact point highlights the moment of ball-paddle interaction. (e) Ball trajectory displays the resulting flight path of the ball.

ABSTRACT

Learning to identify serve spin is a fundamental yet challenging skill for table tennis beginners because the relevant motion cues are subtle and often difficult to observe in traditional training settings. We present an immersive system that reconstructs realistic serve motions from multi-view videos and provides flexible viewpoints, self-directed practice, and visualization tools that reveal essential spatiotemporal cues. The system enables learners to explore serve motions interactively within a virtual environment. A study with novice participants illustrates the promise of this approach for supporting early-stage perceptual learning in table tennis training.

Index Terms: Sports, Visual Design, Immersive Learning

1 INTRODUCTION AND MOTIVATION

During the early phases of a table-tennis rally, the serve and return strongly influence game flow. Skilled servers can disguise different spins with similar motions, making rapid and accurate spin recognition difficult. As a result, identifying serve spin is a fundamental yet challenging skill for beginners. Traditional learning methods, such as demonstrations or online videos, often fail because key cues are subtle, abstract, or constrained by fixed viewpoints [3]. In contrast to traditional methods, an increasing number

of studies have incorporated computer vision and related technologies into table tennis training to enhance skill acquisition. Moreover, prior research demonstrates that visualizing abstract motion information in virtual environments can significantly facilitate the learning of sports skills. Lin et al. [5] reconstructed athlete motions in VR with landing heatmaps and scoring information for tactical analysis. In table-tennis-focused research, Oagaz et al. [6] visualized skeletal models for swing training, and Erwin Wu et al. [9] introduced ball-based cues to teach novices how to return spinning shots. These studies highlight the value of presenting key information through visualization. However, most existing approaches focus on large-scale motion or general motor skills, with limited support for fine-grained cues such as wrist orientation, racket-face angle, and subtle ball trajectories. Our interviews with professional players further indicate that combining these cues is essential for accurate and efficient learning. To address these challenges in traditional training methods or recent training systems, we propose a system that reconstructs the server's motion, paddle orientation, and ball trajectory in a virtual environment and visualizes essential spin-related cues. This environment allows learners to observe serves from any viewpoint and interact with visualization tools that highlight critical motion characteristics.

2 SYSTEM DESIGN GOALS AND IMPLEMENTATION

Our system supports immersive, self-directed training for table tennis serve spin recognition. Based on interviews with professional players, instructional literature [2], and prior research [4], we identified four design goals and developed system components that address key challenges in learning to recognize serve spin type.

Providing Fine-Grained Motion Animation in a Virtual Environment. Although multi-view motion reconstruction methods are now widely available, they primarily focus on whole-body kine-

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matics and often fail to capture the fine wrist details critical for serve spin identification. As a result, directly applying these methods leads to inaccuracies in wrist articulation and paddle orientation. To address this, we reconstruct overall motion using Easy-Mocap [1], adjust wrist angles based on the inferred paddle face, and apply smplify-x [7] hand pose reconstruction to correct wrist pose errors. To further improve the accuracy of paddle motion, we use a unified multi-view 3D reconstruction framework for paddle and ball trajectories. The system applies SAM2 [8] to generate multi-view masks without manual annotations, extracts geometric features through fitting, and identifies consistent correspondences using epipolar constraints. Paddle face orientation and ball position are then reconstructed through stereo matching and triangulation. A dynamic view-pair selection module evaluates available camera pairs and chooses the optimal configuration per frame, improving reconstruction accuracy and temporal stability. Together, these steps enable fine-grained and stable motion animation within the virtual environment.

Linking First-Person View and Multi-View Observations.

Identifying spin types requires fusing multiple sources of information, including paddle motion and the ball's flight path. Yet, obtaining a comprehensive view of subtle details, such as the contact angle between the paddle and the ball, is often hindered by limited viewing angles. To bridge this gap, we introduce an immersive learning environment that allows users to observe serves from arbitrary viewpoints, with the ability to seamlessly transition to an ego-centric (first-person) perspective. This design facilitates the cognitive transfer of multi-view visual information into the first-person experience required for actual competition.

Providing a Self-Learning Pipeline. Traditionally, serve spin recognition relies heavily on expert supervision, which constrains training flexibility. Our system addresses this limitation by facilitating autonomous performance evaluation, enabling learners to engage in continuous, self-directed practice independent of instructor availability. To support this, the system randomly presents serve motions without revealing the spin type, prompting users to self-test their understanding.

Visualizing the Spatial-Temporal Features of Key Strokes.

Serve spin type recognition requires interpreting abstract spatiotemporal cues such as paddle orientation and the moment of contact. Through interviews with professional players, we designed five visualization tools to highlight the essential information needed for understanding serve spin type, as shown in Figure 1. Paddle trajectory (Figure 1(a)) and ghost paddles (Figure 1(b)) help users observe how paddle path relates to the moment of contact. Paddle direction (Figure 1(c)) illustrates the relationship between swing direction and spin generation. The contact point (Figure 1(d)) reveals how the strike location influences spin characteristics. Ball trajectory (Figure 1(e)) allows users to compare how different spins produce distinct flight paths.

Our system combines realistic serve reconstruction, spin visualizations, and multi-view learning into a unified platform, providing beginners with an immersive and practical way to learn serve spin type recognition.

3 EXPERIMENT

We conducted a user study to evaluate the effectiveness of our system features with 51 beginner learners. Participants completed a 90-minute training session, consistent with the duration of a typical table tennis class, and were assigned to one of five learning conditions. The groups were defined as **C1** with four real camera views, **C2** with four fixed 3D views in a virtual environment, **C3** with seven predefined views and a free-view option, **C4** with self-testing added to **C3**, and **C5** with visualization tools added to **C4**.

The study included a pre-test and three evaluation sessions to assess participants' learning progress. In addition to the immediate

Table 1: Accuracy statistics for each condition group in the Pre-test and three post-training tests. Bold indicates the highest score in each group, and underline indicates the second highest.

Group	Pre-test (%)	Test1 (%)	Test2 (%)	Test3 (%)
C1	34.00	49.00	48.00	58.00
C2	35.00	61.00	74.00	<u>77.00</u>
C3	34.00	59.50	65.50	71.50
C4	33.50	<u>69.00</u>	<u>76.50</u>	74.00
C5	35.45	72.73*	80.45*	81.36*

diagnostic evaluations, a delayed post-test was administered one week post-training to assess long-term skill retention. As detailed in Table 1, results indicate that performance improved consistently as additional system features were introduced. The delayed test further indicates that participants with access to additional viewpoints retained serve-spin knowledge better, suggesting that multi-view learning (**C3-C5**) supports both immediate learning and longer-term retention.

4 CONCLUSION AND FUTURE WORK

This work presents an immersive training system that reconstructs realistic serve motions in a virtual environment and visualizes key cues relevant to learning table tennis serve spin recognition. Our user study shows that combining multi-view learning, self-learning mechanisms, and spin-related visualizations can support beginners in developing their understanding of serve spin cues. These findings suggest that virtual environments can play a constructive role in supporting perceptual-motor learning for complex sports tasks.

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